Week_1: Short Notes

Lecture 1: What is Machine Learning?

- **Definition**: ML is a method where computers learn from data instead of being explicitly programmed.
- **Basic Idea**:
    - Programming → Humans write rules
    - ML → Humans provide data + idea, machine learns rules
- **When to Use ML**:
    - Task too large, fast, or costly for humans
    - Rules are complex or unknown
- **Requirements**: Large data + basic understanding of problem
- **Examples**:
    - Password verification → programming
    - Face detection → ML
    - Weather prediction → ML
- **Applications**: Spam detection, recommendations, voice assistants, robotics, games, marketing
- **Summary**: ML = learning from data, used when rules are unclear or tasks complex.

Lecture 2: Data, Models, and ML Task

- **Data**: Collection of vectors (e.g., house → (3, 9, 1.9, 5.0))
- **Metadata**: Meaning of numbers (rooms, area, distance, price)
- **Models**: Simplified versions of reality, useful for prediction
- **Types of Models**:
    - Predictive → regression (continuous), classification (categories)
    - Probabilistic → likelihood estimation
- **Learning Algorithms**: Convert data into models by finding best parameters
- **Task Flow**: Human gives data + structure → algorithm learns → model predicts.

Lecture 3: Supervised Learning – Regression

- **Notation**:
    - Real numbers R, vectors R^d
    - Indexing:
        - x_i: i-th data point (vector)
        - x_j: j-th value inside a vector
        - x_{ij}: j-th value of i-th data point
- **Regression**: Learn function $f(x): R^d \to R$
- **Linear Model**: $f(x) = w^T x + b = w_1x_1 + w_2x_2 + \cdots + w_dx_d + b$
- **Loss Function**: Squared loss → $(y-f(x))^2$
- **Goal**: Minimize loss, find realistic patterns.
- **Summary**: Regression predicts numeric values, best model = minimum loss.

Lecture 4: Supervised Learning – Classification

- **Definition**: Predict discrete labels (+1, −1)
- **Model**: $f(x) = \text{sign}(w^T x+b)$
- **Loss**: Misclassification loss = fraction of wrong predictions
- **Data Splitting**: Training, validation, test sets
- **Summary**: Classification predicts categories, performance measured by misclassification, best model generalizes well.

Lecture 5: Unsupervised Learning – Dimensionality Reduction

- **Definition**: Learning without labels
- **Goal**: Understand structure, group data, compress data
- **Encoder/Decoder**:
  - $f:R^D \rightarrow R^{d'}$
  - $g:R^{d'} \rightarrow R^D$
- **Loss Function**: $1/n\sum\|g(f(xi))-xi\|^2$
- **Applications**: Marketing, data compression
- **Summary**: Reduce dimensionality, minimize reconstruction error.

Lecture 6: Unsupervised Learning – Density Estimation

- **Definition**: Learn probability distribution $P(x)$
- **Constraints**:
  - Discrete: $\sum P(x)=1$
  - Continuous: $\int P(x)dx=1$
- **Loss Function**: Negative log likelihood $L = 1/n\sum -\log P(xi)$
- **Applications**: Tweet generation, Gaussian Mixture Models
- **Summary**: Density estimation assigns high probability to real data, minimizes loss.